

6260 Final Project Report

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Introduction

Criminal Justice outcomes are often studied as whether contact has occurred, but this can hide many meaningful differences when it comes to seriousness. Individuals may differ not only in whether they have contact but also severity. Due to early life socioeconomic and family conditions affecting adult trajectories, examining severity rather than simply contact may provide a fuller picture of how disadvantage accumulates across life courses. In hopes of gaining a more comprehensive view of trajectory, our study aims to capture the multidimensional nature of upbringing, involving factors such as education, housing instability, and family structure.

This perspective is supported by prior literature linking criminal justice involvement to broader systems of inequality. Turney and Wakefield (2019) argue that criminal justice contact is a central mechanism through which inequality is produced and maintained, extending beyond incarceration itself into families, schools, neighborhoods, and other institutions. Their work suggests that criminal justice outcomes are concentrated among disadvantaged populations and that early-life disadvantages may shape later justice trajectories. Similarly, Côté-Lussier, Hohl, and David (2020) describe disadvantage and criminal justice as mutually reinforcing: disadvantage can increase criminal justice involvement, while criminal justice involvement can deepen later inequality. Their review is especially useful for this project because it frames criminal justice outcomes as part of a broader cumulative process rather than an isolated event.

In addition, Slade et al. (2008) provide a more direct life-course example by showing that, among disadvantaged urban young men, earlier onset of substance use disorder is associated with greater adult incarceration and broader criminal justice involvement. Their findings support the broader idea that early risk exposures may matter not only for whether criminal justice contact occurs, but for how serious later justice outcomes become. Taken together, these studies motivate the present project’s focus on early-life disadvantage and adult criminal-history severity.

In hopes of addressing the need for a more detailed outcome, we design our own variable to more accurately describe severity, which synthesizes time and crime type to have a comprehensive detail of severity. This approach treats severity as an ordinal quantity rather than a continuous one, allowing the analysis to preserve ranking without requiring equal spacing between severity levels. Because of this structure, a cumulative logit model provides a natural framework for assessing how early-life predictors are associated with movement into higher severity categories.

The analysis is intended as a predictive and explanatory study of association, rather than a causal claim. The paper proceeds as follows. First, the data source and variable construction are described. Next, the ordinal severity response is developed and justified. Exploratory data analysis is then used to examine the distribution of the response and the candidate predictors, as well as assumptions relevant to ordinal modeling. The cumulative logit framework is then introduced and interpreted. Finally, limitations, future improvements,

and policy implications are discussed.

Data

The data used in this study is the Survey of Prison Inmates (SPI), United States, 2016 (ICPSR 37692) Data Set for Restricted Use.

In trying to understand an individual’s trajectories, efforts should be made to try to look at many facets of upbringing. The Bureau of Justice Statistics (BJS) attempts to do this, with the final dataset containing inmate records, intake forms, surveys, questionnaires, screenings, etc. with a total of 2259 columns. This survey was conducted so that policymakers could address shortcomings of correctional institutions. The sample selection was done with a 2 stage cluster probability sample, where The first stage consisted of a random sample of prisons. The second-stage consisted of a random sample of prisoners held in the selected prisons. 20,000 inmates were chosen to represent about 1.5 million inmates within the prison system. Data was collected using Computer-Assisted Personal Interviewing to assist with human and data entry error with a relatively high response rate (70%).

Many of the variables were suppressed due to privacy reasons and therefore had to be removed because they were unusable. Additionally, all categorical variables had extra descriptions, i.e. “(1) 1 = Yes” as an observation, and therefore must be flattened. Furthermore, many observations were missing for various reasons with codes to specify why, but for our purposes, we flattened them all to “0” or NA as the reason for missing was not relevant to this project.

Questions regarding personal feelings on past wrongdoings may lead to response biases as inmates may feel they want to lie to “look good.” Inmates were assured that answers were completely anonymous and that their identity was protected, to reduce bias. The final model is comprised of

Variable Name	Description	Data Type
Years of Education	Years of Education Completed	Ordinal (5 Classes)
Children	Does offender have children	Binary (Y/N)
Foster Home	Was offender part of foster home before 18	Binary (Y/N)
Homeless	Was offender homeless before 18	Binary (Y/N)
Welfare	Welfare/public assistance before age 18	Binary (Y/N)
Father/Mother/Sibling	Offender’s [relative] ever been sentenced/served time	Binary (Y/N)
DrinkAge/SmokeAge	Age when first drank alcoholic beverage/started smoking fairly regularly	Numeric (Input age)

The primary response variable in this analysis is adult criminal-history severity, operationalized through a constructed ordinal measure called TotalScore.

This variable was created to move beyond a simple indicator of criminal justice contact and instead capture meaningful differences in the seriousness of inmates’ criminal histories. A binary outcome such as incarcerated

versus not incarcerated would not distinguish between individuals with very different offense profiles and sentencing burdens. Because the goal of the project is to study variation in the severity of adult criminal history, a broader outcome measure was needed.

TotalScore is comprised of 2 individual scores: CrimeScore, which is based on Offense Type, and TimeScore, which is based on max cumulative sentence length.

$$TotalScore_i = CrimeScore_i + TimeScore_i$$

Crime type captures the qualitative seriousness of the underlying offense, while sentence length reflects the magnitude of punishment. These components were added together to create an ordered severity score, with higher values representing more severe criminal histories overall.

$$TotalTime_i = \frac{365(MaxYears_i) + 30(MaxMonths_i) + MaxDays_i}{30.44}$$

To construct the time-based component, sentence length was first converted into a common monthly scale

$$TimeScore_i = \begin{cases} 1 & \text{if } Tot_time_i < 12 \\ 2 & \text{if } 12 \leq Tot_time_i < 36 \\ 3 & \text{if } 36 \leq Tot_time_i < 84 \\ 4 & \text{if } 84 \leq Tot_time_i < 180 \\ 5 & \text{if } 180 \leq Tot_time_i < 360 \\ 6 & \text{if } Tot_time_i \geq 360 \end{cases}$$

The sentence-length categories were defined using substantively meaningful ranges rather than purely data-driven cutoffs. These ranges distinguish short-term sentences from progressively more serious and long-term sentencing exposure, allowing the time component to reflect punishment severity in a way that is more interpretable than raw sentence length alone.

Offense type categories were assigned an ordinal CrimeScore, where higher values correspond to more serious crimes within the scoring system

$$CrimeScore_i = \begin{cases} 5 & \text{if offense tpe is Violent} \\ 4 & \text{if offense tpe is Property} \\ 3 & \text{if offense tpe is Drug} \\ 2 & \text{if offense tpe is Public Order} \\ 1 & \text{if offense tpe is Other} \end{cases}$$

This produces an approximate total sentence duration in months. The variable was then converted into an ordinal TimeScore using substantively meaningful sentencing ranges:

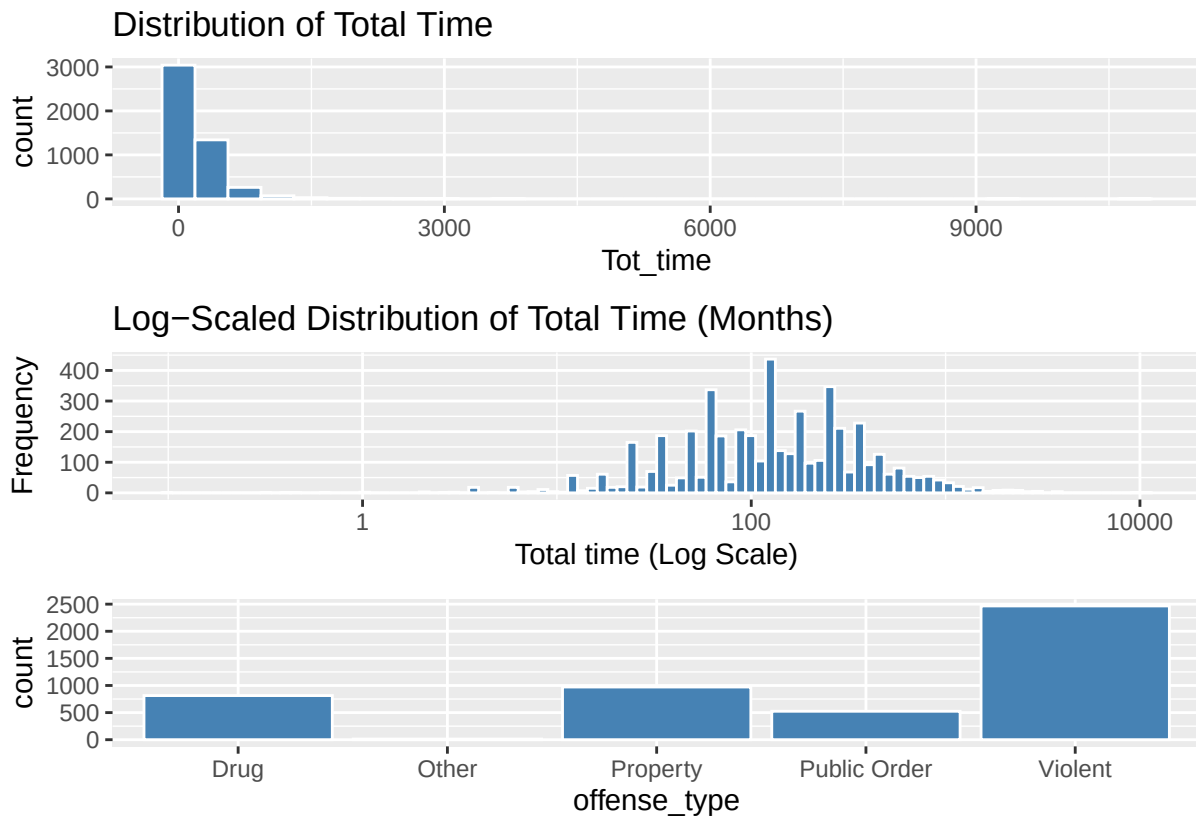
The use of a composite response is supported by exploratory analysis. Total time alone was highly right-

skewed, indicating that sentence length by itself would be difficult to interpret and overly influenced by a relatively small number of extreme cases. At the same time, offense categories showed meaningful variation, suggesting that type of crime also contributes important information about severity. For this reason, neither offense type nor sentence length alone was considered sufficient. The combined score was intended to better reflect the multidimensional nature of criminal-history severity.

This response should be interpreted as an ordinal severity index, not as a precise continuous measurement. That is, higher values indicate greater criminal-history severity, but the difference between adjacent score levels should not be interpreted as equal in magnitude. Because the measure is researcher-constructed, it is also subject to limitations: the final score depends on coding choices for offense seriousness and sentence-length categories, and alternative scoring schemes could produce somewhat different results. Nevertheless, the composite outcome provides a more substantively meaningful measure of severity than either component in isolation.

Exploratory Data Analysis

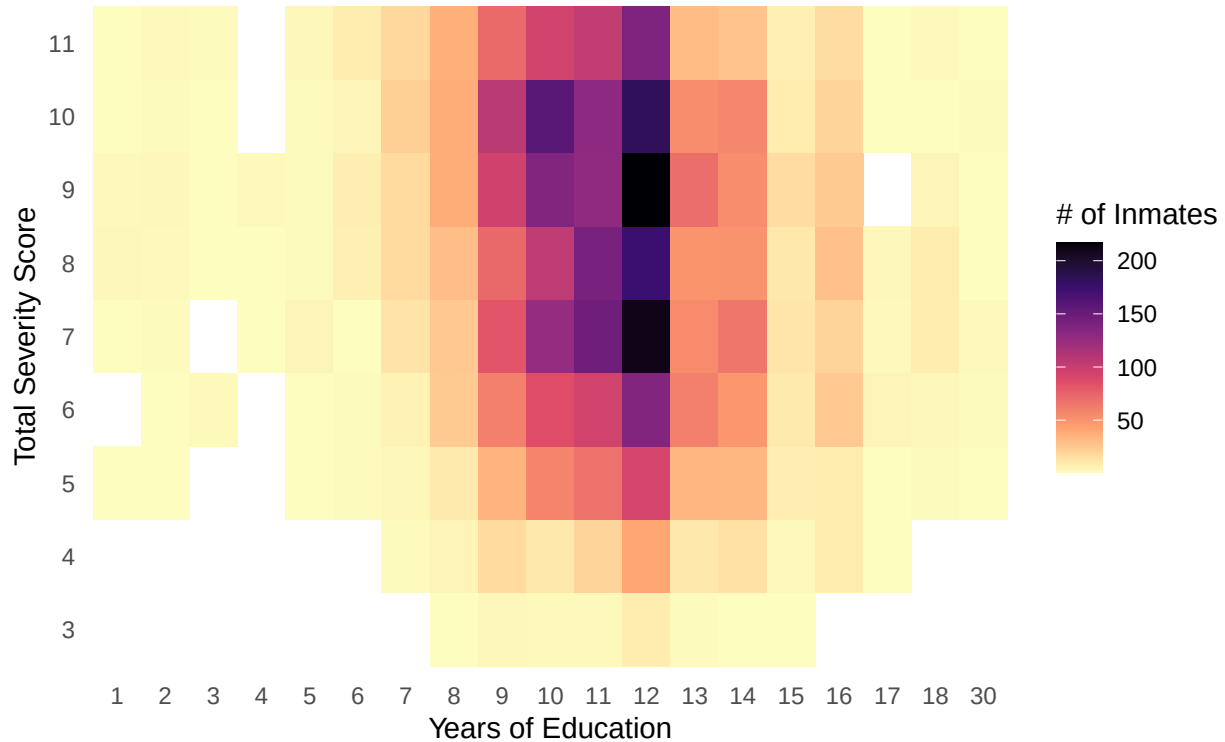
We conducted exploratory data analysis to further our understanding of the variables in our data set. Using data visualizations of the distribution of variables and correlation matrices allowed us to see the range of outcomes for prison inmates.



The above plots show the distribution of the components of our response variable. The time variable is heavily right skewed. That means a small number of individuals have very long sentences, while most observations are concentrated at lower values. This makes raw sentence length hard to use directly, because a few extreme cases can dominate the scale. When we apply a log transformation, the distribution becomes much more regular, which tells us that time is meaningful, but still only one dimension of severity. At the same time, offense frequencies vary substantially across categories, with violent offenses most common, indicating that qualitative differences in offense seriousness are also important. These patterns support constructing a composite ordinal response variable that combines sentence burden and offense seriousness into a broader measure of severity.

Concentration of Inmate Profiles

Education vs. TotalScore

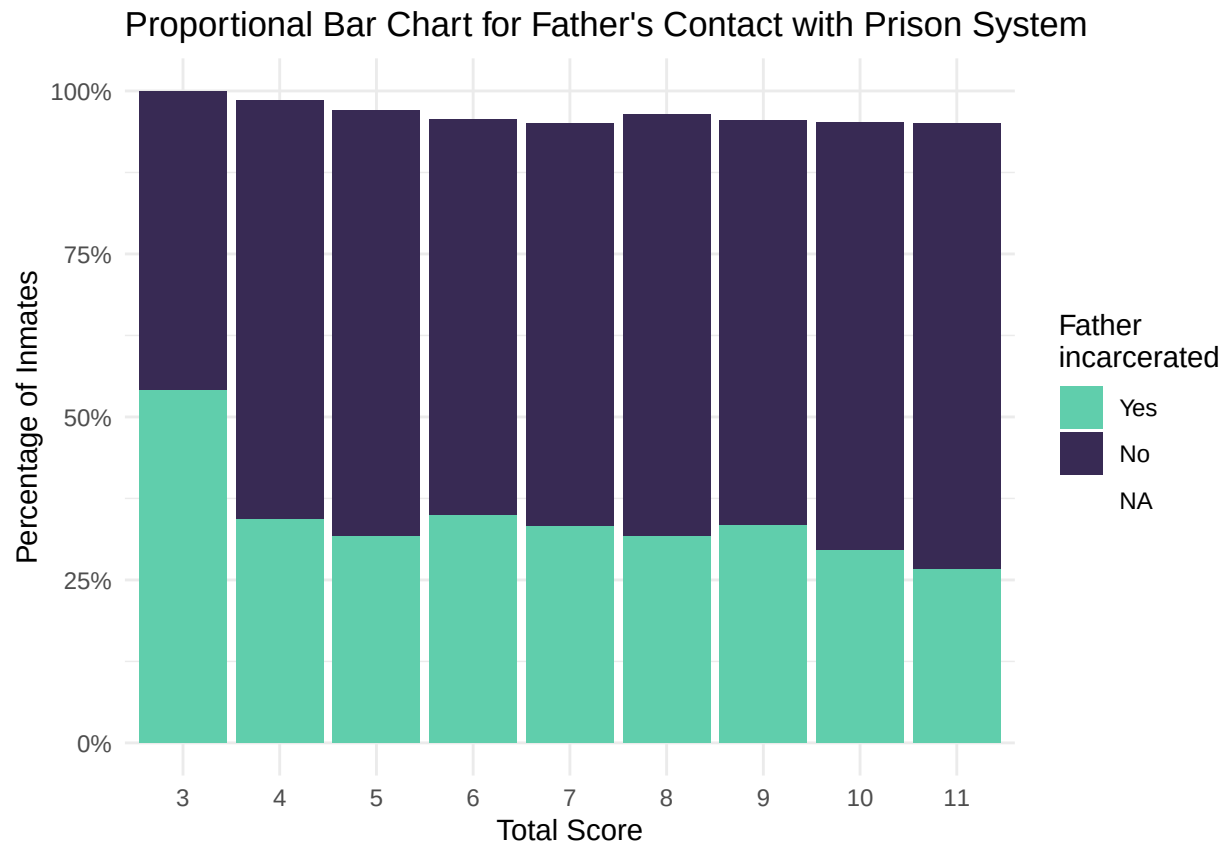


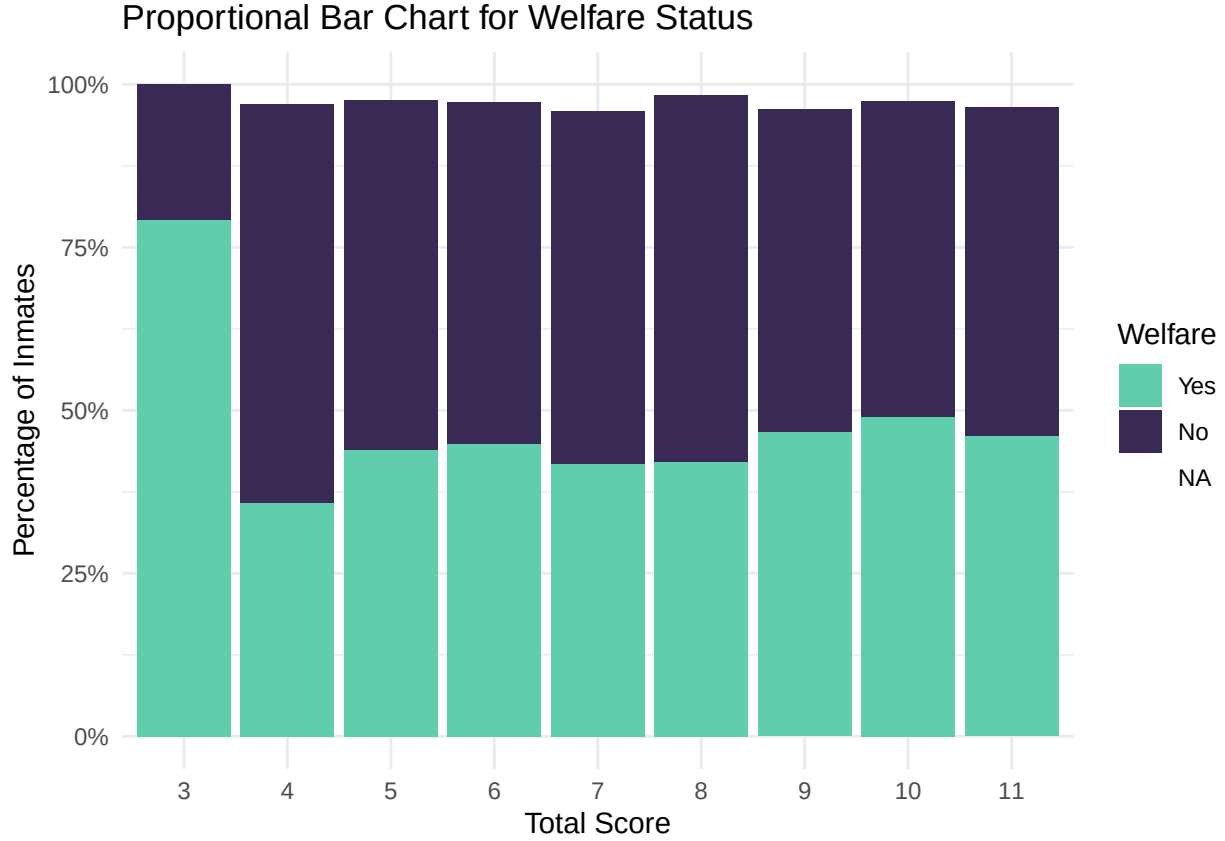
The heatmap shows that inmate profiles are most concentrated among individuals with roughly 9–12 years of education and moderate-to-high severity scores. While this indicates where the sample is densest, it does not by itself show a clear monotonic relationship between education and severity, suggesting that additional covariates may be needed to explain variation in criminal-history severity. For this reason, years of education was converted into an ordinal categorical variable with five classes. This decision was made for 2 reasons. Substantively, education is often interpreted through milestones such as incomplete schooling, high school completion, and education beyond high school, rather than through the assumption that each additional year has the same effect. Statistically, grouping education reduces sensitivity to sparse counts at very low and very high values and avoids imposing a strictly linear structure when the exploratory plots did not support one clearly.

Variable	Missing n	Missing %	Mean	Median	SD	Min	Max
DrinkAge	405	8.46	14.15	14	4.06	1	61
SmokeAge	1098	22.92	15.06	15	4.42	0	47

The age-of-initiation variables suggest that, among nonmissing observations, first alcohol use and first regular smoking generally occurred in the mid-teen years. DrinkAge had a mean of 14.03 years and median of 14, while SmokeAge had a mean of 15.02 years and median of 15, indicating that alcohol initiation occurred

slightly earlier on average than smoking initiation. Missingness was higher for SmokeAge (22.71%) than for DrinkAge (7.77%), which may limit comparability across the two predictors.





Proportional bar charts for welfare exposure and father’s contact with the prison system suggest that the composition of these early-life disadvantage measures varies across levels of TotalScore, but neither predictor exhibits a clear monotonic bivariate relationship with criminal-history severity. In both cases, the proportion of “Yes” responses fluctuates across severity categories rather than steadily increasing or decreasing. This suggests that any relationship between these early-life factors and adult criminal-history severity may be modest, nonlinear, or dependent on additional covariates. Additionally, differences at the lowest severity levels should be interpreted with a grain of salt because those categories contain fewer observations (24).

Analysis Proposal

In that we are working with an ordinal response variable, as described in the setup of this question, we will be using a cumulative logit model to model this response of the total score comprised of sentence time and severity of the crime. The cumulative logit model itself is expressed as

$$\text{logit}[P(Y \leq j|x)] = \alpha_j + \beta^T x \text{ where } \text{logit} = \frac{p}{1-p}$$

In words, this model calculates the probability that a prisoner falls in a category at or below the one predicted given a set of predictors. From this model, there are some assumptions that arise that must be looked at in order for our results to be valid. For one, as has been discussed, the response must be an ordinal categorical

variable in nature. This must be the case because if not, then looking at the probability that a response would fall in some category or less would not make sense as nominal variables, the other type of categorical variables, do not have a specific ordering to them. Thus, using a less than or equal to sign would not make sense for nominal variables, explaining why an ordinal response must be used.

The second and last assumption of this model is the proportional odds assumption. This assumption states that the effect of any predictor on the odds of a response being in or below a certain category is the same regardless of which category is being considered. Even though each cumulative logit has its own intercept (α_j), the coefficients (β) must be the same throughout each category. This assumption is very crucial to ensure relationships do not differ between different categories, as the model states this must be the case.

Furthermore, instead of the logit link, we will consider looking at other link functions to see if these link functions provide better fit to the data. These other link functions that will be tested are the probit and complementary log log function. The probit link function, Φ^{-1} models the response using a standard normal CDF. The complementary log log link function, $\log(-\log(1 - p))$ is an asymmetric link function, which is different from probit and logit link functions as these are symmetric.

Analysis

In implementing the proposed analysis, we used the MASS package in R in order to get the results. As mentioned, we are using the total score variable as the response, and all the variables listed in the table in the data section will be used to model the response. Furthermore, we will use an 80/20 train and test split in order to be able to test the results of our model to further evaluate how the data reacts to the model given. The results from the cumulative logit model can be seen below:

```
## Call:
## polr(formula = TotalScore ~ as.factor(edu_group) + as.factor(Children) +
##       as.factor(`Foster Home`) + as.factor(Homeless) + SmokeAge +
##       DrinkAge + as.factor(Welfare) + as.factor(Mother) + as.factor(Father),
##       data = train_data, Hess = TRUE)
##
## Coefficients:
##                                     Value Std. Error  t value
## as.factor(edu_group)College+      -0.292223   0.183948 -1.58862
## as.factor(edu_group)HS/GED        -0.231836   0.085650 -2.70677
## as.factor(edu_group)Other         -1.234481   0.738466 -1.67168
## as.factor(edu_group)Some college -0.302123   0.098432 -3.06935
## as.factor(Children)2              0.237349   0.075998  3.12308
```

```

## as.factor(`Foster Home`)2      -0.001724    0.086819 -0.01986
## as.factor(Homeless)2            -0.102856    0.101427 -1.01409
## SmokeAge                        0.005371    0.008433  0.63688
## DrinkAge                       -0.032821    0.009691 -3.38685
## as.factor(Welfare)2            -0.139226    0.074436 -1.87042
## as.factor(Mother)2             0.176580    0.102368  1.72496
## as.factor(Father)2             0.234293    0.079488  2.94753
##
## Intercepts:
##      Value      Std. Error t value
## 3|4      -5.2253    0.2904   -17.9950
## 4|5      -3.6292    0.2067   -17.5609
## 5|6      -2.2934    0.1871   -12.2604
## 6|7      -1.3961    0.1825    -7.6496
## 7|8      -0.6208    0.1807    -3.4347
## 8|9       0.0088    0.1804     0.0491
## 9|10      0.7867    0.1811     4.3450
## 10|11     1.9359    0.1867    10.3702
##
## Residual Deviance: 10410.56
## AIC: 10450.56

```

To start off, it must be pointed out that categorical factors that are coded in as a two is a no. From these results, it can be seen that level of education obtained, having children or not, and drinking age seem to be the most important predictors given their test statistic values. Even though p-values are not calculated, and we will not do since we are not looking for significance, these results allow us to see which factors influence a person's tendency to commit a more heinous crime given a set of predictors. In talking about the education variable, the base level for this is less than high school. The negative coefficients for every variable here indicates that there is a negative association between level of crime committed and education level obtained, meaning higher education is associated with a less severe crime committed. Next, in looking at the children variable, there can be seen a positive coefficient, indicating that having kids is associated with a less severe crime since having a kid is the baseline. Lastly, in pointing out the drinking age predictor, this has the highest test statistic value in terms of magnitude. Its negative coefficient means that starting to drink alcohol at a younger age is associated with a more severe crime. In the conclusion, we will explain some interesting reasons for why this may be the case.

In further wanting to assess this model, it is important to check the assumptions to ensure conclusions we

draw are not invalid. In order to do this, we will use the nominal test function from the ordinal package. What this function does is it does a likelihood ratio test comparing a model assuming the assumption of proportional odds is met versus a model that assumes the odds change from category to category. The p-values will then indicate whether the more flexible model, the one not assuming the proportional odds assumption, is a better fit or not. Low p-values will suggest this more flexible model is a better fit, thus indicating this assumption may not be met. The results from doing this can be seen below :

```
## Tests of nominal effects
##
## formula: as.factor(TotalScore) ~ edu_group + Children + FosterHome + Homeless + SmokeAge + DrinkAge
##           Df  logLik   AIC   LRT Pr(>Chi)
## <none>      -5205.3 10451
## edu_group
## Children    7 -5204.1 10462 2.3793  0.9359
## FosterHome  7 -5201.0 10456 8.4806  0.2921
## Homeless    7 -5202.8 10460 4.9204  0.6697
## SmokeAge    7 -5203.2 10460 4.0659  0.7722
## DrinkAge    7 -5202.8 10460 5.0289  0.6564
## Welfare
## Mother      7 -5202.6 10459 5.4369  0.6068
## Father      7 -5203.2 10460 4.2090  0.7554
```

From the output, there can be many high p-values seen. This is good as it indicates the assumption of proportional odds is met. It is also important to point out that the education level do not have an associated p-value. This is due to the fact that there is an identifiability issue, as in the class with a response total score of 3, there is nobody with education level of other. This makes it so the maximum likelihood estimate for the education variable cannot be determined, as the model is unable to calculate the odds of moving between categories for a group that has no observations in one of those categories. Since this is the only variable we cannot determine whether the proportional odds assumption can be met for it, and the rest of the variables have very insignificant p-values, we will assume this assumption is met for our analysis.

For the next part of our analysis, we would like to look at predictions from our test set to understand how the predictions will look. From this, the confusion matrix can be seen below :

```
##           Predicted
## Actual  3  4  5  6  7  8  9 10 11
##      3   0  0  0  0  2  0  1  1  0
##      4   0  0  0  0 13  0  6  2  0
##      5   0  0  0  0 29  0 18  8  0
```

```
##      6      0      0      0      0 39      0 33      5      0
##      7      0      0      0      0 72      0 49     14      0
##      8      0      0      0      0 42      0 38     12      0
##      9      0      0      1      1 28      0 44     16      0
##     10      0      0      0      1 42      0 32     18      0
##     11      0      0      0      0 24      0 36     20      0
```

The confusion matrix shows a majority of 7's, 9's, and 10's predicted, with an overall accuracy of about 21 percent. Reasons for this look to be that our model is just defaulting to some of the highest observed classes.

```
##
##      3      4      5      6      7      8      9     10     11
##    23     93    278    414    569    490    529    500    339
```

As seen above, the categories 7,9, and 10 are the most observed classes. Predictions like this usually occur when the model does not have a lot of good information, so it predicts the most common classes very often. In order to further investigate this, we would like to see what the AIC and residual deviance of the null model, a model with no predictors, is to see if our model truly is good or not.

```
## Call:
## polr(formula = TotalScore ~ 1, data = train_data, Hess = TRUE)
##
## No coefficients
##
## Intercepts:
##      Value      Std. Error t value
## 3|4      -4.9068      0.2303   -21.3028
## 4|5      -3.3122      0.1067   -31.0317
## 5|6      -1.9800      0.0602   -32.8869
## 6|7      -1.0905      0.0453   -24.0699
## 7|8      -0.3259      0.0398    -8.1804
## 8|9       0.2942      0.0397     7.4036
## 9|10      1.0598      0.0450    23.5692
## 10|11     2.1963      0.0655    33.5308
##
## Residual Deviance: 10465.84
## AIC: 10481.84
```

From this, it can be seen that the AIC's and residual deviances drop a very small amount. The AIC drops

from 10481.84 to 10450.56 in going from the null model to the full model, and the residual deviance from 10465.84 to 10410.56. This is a very miniscule difference, suggesting our model does not explain much of the variation in the response. Because of this, we would like to test if different link functions may create a better fit.

The probit link function attempt can be seen below:

```
probit_model <- polr(TotalScore ~ as.factor(edu_group)+as.factor(Children) +
                    as.factor(`Foster Home`) + as.factor(Homeless) +
                    SmokeAge + DrinkAge + as.factor(Welfare) + as.factor(Mother) +
                    as.factor(Father), data = train_data, method = "probit", Hess = TRUE)
summary(probit_model)
```

Call:

```
## polr(formula = TotalScore ~ as.factor(edu_group) + as.factor(Children) +
##      as.factor(`Foster Home`) + as.factor(Homeless) + SmokeAge +
##      DrinkAge + as.factor(Welfare) + as.factor(Mother) + as.factor(Father),
##      data = train_data, Hess = TRUE, method = "probit")
##
```

Coefficients:

	Value	Std. Error	t value
## as.factor(edu_group)College+	-0.147724	0.110590	-1.3358
## as.factor(edu_group)HS/GED	-0.135218	0.049657	-2.7231
## as.factor(edu_group)Other	-0.708081	0.454134	-1.5592
## as.factor(edu_group)Some college	-0.180126	0.056873	-3.1672
## as.factor(Children)2	0.134824	0.044192	3.0509
## as.factor(`Foster Home`)2	-0.003177	0.050911	-0.0624
## as.factor(Homeless)2	-0.057456	0.058515	-0.9819
## SmokeAge	0.003896	0.004949	0.7872
## DrinkAge	-0.019684	0.005622	-3.5015
## as.factor(Welfare)2	-0.065138	0.043195	-1.5080
## as.factor(Mother)2	0.102542	0.059874	1.7126
## as.factor(Father)2	0.133494	0.046389	2.8777

##

Intercepts:

	Value	Std. Error	t value
## 3 4	-2.6169	0.1316	-19.8928
## 4 5	-1.9825	0.1125	-17.6176

```
## 5|6      -1.3379   0.1074  -12.4542
## 6|7      -0.8350   0.1060   -7.8797
## 7|8      -0.3641   0.1053   -3.4564
## 8|9       0.0283   0.1052    0.2692
## 9|10     0.5020   0.1054    4.7612
## 10|11    1.1396   0.1071   10.6414
##
## Residual Deviance: 10412.50
## AIC: 10452.50
```

The results from the complementary log log link function can be seen below:

```
cloglog_model <- polr(TotalScore ~ as.factor(edu_group)+as.factor(Children) +
                        as.factor(`Foster Home`) + as.factor(Homeless) +
                        SmokeAge + DrinkAge + as.factor(Welfare) + as.factor(Mother) +
                        as.factor(Father), data = train_data, method = "cloglog", Hess = TRUE)
summary(cloglog_model)
```

```
## Call:
## polr(formula = TotalScore ~ as.factor(edu_group) + as.factor(Children) +
##      as.factor(`Foster Home`) + as.factor(Homeless) + SmokeAge +
##      DrinkAge + as.factor(Welfare) + as.factor(Mother) + as.factor(Father),
##      data = train_data, Hess = TRUE, method = "cloglog")
##
```

Coefficients:

	Value	Std. Error	t value
## as.factor(edu_group)College+	-0.200548	0.115261	-1.7399
## as.factor(edu_group)HS/GED	-0.110835	0.051929	-2.1344
## as.factor(edu_group)Other	-0.933656	0.458691	-2.0355
## as.factor(edu_group)Some college	-0.161159	0.059374	-2.7143
## as.factor(Children)2	0.134236	0.046542	2.8842
## as.factor(`Foster Home`)2	0.049857	0.052400	0.9515
## as.factor(Homeless)2	-0.090128	0.061043	-1.4765
## SmokeAge	0.002442	0.005078	0.4809
## DrinkAge	-0.020051	0.005428	-3.6942
## as.factor(Welfare)2	-0.090742	0.045532	-1.9929
## as.factor(Mother)2	0.102535	0.062541	1.6395
## as.factor(Father)2	0.149351	0.049062	3.0442


```
##
## Intercepts:
##      Value      Std. Error t value
## 3|4      -5.0904    0.2521   -20.1945
## 4|5      -3.5092    0.1478   -23.7390
## 5|6      -2.2237    0.1186   -18.7514
## 6|7      -1.4145    0.1114   -12.7001
## 7|8      -0.7817    0.1085    -7.2010
## 8|9      -0.3289    0.1074    -3.0613
## 9|10      0.1449    0.1068     1.3576
## 10|11     0.6826    0.1067     6.3978
##
## Residual Deviance: 10411.08
## AIC: 10451.08
```

As seen, these link functions do not help the fit much, if at all. The AIC's and residual deviances of the models just tested are all higher than those from the logit link function, but not by a significant margin.

Conclusion